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ORIGINAL PAPER

RSA-YOLO: an improved YOLOv8-based model for tomato leaf disease detection

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Abstract
Currently, tomato leaf disease detection faces critical challenges, including ambiguous multi-scale target recognition, strong background interference, and limited robustness under varying lighting conditions. Although recent state-of-the-art (SOTA) YOLO-based detectors have achieved notable progress, their performance remains constrained under complex agricultural scenarios, mainly due to limited receptive-field adaptability, insufficient multi-scale feature fusion, and weak scale awareness. To address these limitations, an efficient tomato leaf disease detection model based on YOLOv8n, termed RSA-YOLO, is proposed. A Receptive-Field Attention (RFA) mechanism is introduced into the backbone to enable dynamic receptive-field adaptation and improve feature discriminability. Meanwhile, the Spatial Pyramid Pooling with Efficient Layer Aggregation Network (SPPELAN) is employed to strengthen multi-scale contextual representation, and an Adaptive Spatial Feature Fusion (ASFF) mechanism is incorporated into the detection head to enhance feature scale invariance. Experimental results show that RSA-YOLO consistently outperforms the original YOLOv8n, achieving improvements of 1.6% in Precision, 2.6% in mAP@0.5, and 2.4% in F1-score. Compared with YOLOv3-tiny, YOLOv5n, YOLOv6n, YOLOv9C, YOLOv11n, YOLOv12n, and YOLOv13n, RSA-YOLO achieves mAP@0.5 gains of 6.8%, 4.1%, 4.5%, 2.4%, 2.6%, 3.5%, and 1.9%, respectively. These results demonstrate that RSA-YOLO improves detection accuracy over existing YOLO-based detectors while maintaining a moderate and controllable computational cost (11 GFLOPs) and model size (4.61M parameters), supporting its practical applicability in agricultural disease monitoring scenarios.

Keywords Tomato leaf disease · Object detection · YOLOv8n · RFA · ASFF

1 Introduction

Tomatoes, a globally important cash crop, are typically cultivated in warm and humid environments that are highly susceptible to pathogens and pests. Common tomato leaf diseases, including septoria leaf spot, late blight, and leaf miner infestations, can significantly reduce tomato yield and quality, leading to substantial economic losses for farmers. [1]. Currently, the prevention and control of tomato leaf diseases and pests face challenges due to their diversity and unpredictable outbreaks [2]. Although various chemical control methods are available, curative treatments often incur higher costs than preventive measures. Furthermore, the excessive use of chemical pesticides not only destabilizes agricultural ecosystems but also promotes pest resistance, thereby increasing control expenses [3]. Therefore, early detection of abnormalities during the tomato growth phase, coupled with timely management of diseases and pests, can enhance both yield and quality while reducing pesticide dependence. Consequently, timely detection of tomato leaf diseases at an early stage is critical [4].

In the field of crop disease and pest detection, traditional methods predominantly rely on manual judgment, which is both inefficient and prone to subjective bias, leading to low accuracy and inconsistent results [5]. As the scale of agricultural production expands and geographic diversity increases, these traditional methods struggle to adapt to the complex and dynamic nature of modern agricultural environments. This is largely due to their limited monitoring coverage and inadequate real-time responsiveness [6]. Specifically, in detecting

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early-stage and complex diseases, traditional methods often yield unsatisfactory results.

In recent years, machine learning techniques have been increasingly applied to disease detection to overcome the limitations of traditional manual methods. Madhav et al. [7] utilized Support Vector Machine (SVM) algorithms to classify diseases and control pesticide residues, following the segmentation and processing of fruit images using image processing techniques. Bera et al. [8] developed PND-Net, which integrated Graph Convolutional Network (GCN) with Convolutional Neural Network (CNN) and captured multi-scale features through spatial pyramid pooling. The performance improvements in plant nutrient deficiency and disease classification were validated on multiple laboratory datasets. Prabu et al. [9] proposed an enhanced SVM model based on an arithmetic optimization algorithm (BSVM-AOA), which was combined with a vector-valued active contour model for image segmentation and utilized a gray-level co-occurrence matrix for feature extraction, achieving high-precision detection of plant leaf diseases. Wu et al. [10] improved the ResNet model for strawberry disease identification by strategically embedding two Squeeze-and-Excitation (SE) blocks, which increased recognition accuracy while maintaining low computational costs. However, these machine learning methods are only suitable for single disease recognition in standard laboratory images and cannot identify the specific location of the disease. Iqbal et al. [11] reported similar findings: in small-scale image datasets, some learning algorithms (e.g., AdaGrad) achieved higher accuracy, faster training, and lower memory usage. However, these methods are mainly effective under controlled conditions and may be limited in more complex or large-scale tasks, highlighting the necessity of deep learning-based detection frameworks.

With the significant advancements in deep learning, target detection methods based on Convolutional Neural Networks (CNNs) have gradually become mainstream, leading to two primary technical approaches: two-stage and single-stage detection models. Common two-stage detection models include R-CNN [12], Faster R-CNN [13], and VFNet [14], among others. Using Faster R-CNN as a representative example, this approach achieves superior detection accuracy by generating candidate regions in the first stage and performing fine classification and regression in the second stage. However, it suffers from inherent drawbacks, such as high computational complexity and slow inference speed.

In contrast, single-stage detectors, represented by YOLO [15, 16] and SSD [17], significantly improve efficiency by integrating the target localization and classification tasks into a single, fully convolutional network. For instance, Ullah et al. [18] proposed EffiMob-Net for tomato leaf disease identification. This model enhances feature extraction by fusing EfficientNetB3 and MobileNet, mitigates overfitting through regularization, dropout, and batch normalization, and optimizes performance via hyperparameter tuning. To improve detection of small disease targets, Zhang et al. [19] incorporated an EfficientNet backbone into YOLOv8. This lightweight model achieves higher accuracy with fewer parameters than the original YOLOv8, providing an effective and efficient solution for detecting subtle lesions. Similarly, Chen et al. [20] improved YOLOv8n by integrating a C2f-DynamicConv module for adaptive feature extraction and a SimAM attention mechanism to focus on key regions. In addition, they applied Dysample upsampling to enhance feature map quality and employed GIoU loss to refine bounding box regression, collectively improving detection accuracy, localization precision, and computational efficiency. In complex agricultural environments, Miao et al. [21] proposed SerpensGate-YOLOv8, which incorporates DySnakeConv within the C2F module to capture intricate features, employs the SPPELAN module for enhanced feature fusion, and introduces a Super Token Attention (STA) mechanism for global feature modeling. These improvements enable the model to maintain superior detection performance under variable lighting and occlusion conditions. Moreover, Qiang et al. [22] developed a dual-backbone SSD network by combining ResNet50 and VGG16 in parallel. By leveraging the complementary advantages of deep semantic features and shallow detail information, their model significantly enhances localization accuracy for citrus diseases and pests. These ground-based improvements provide a foundation for the development of lightweight models suitable for UAV and edge-device deployment.

While ground-based improvements enhance detection accuracy, deploying models on UAVs or edge devices requires lightweight architectures to meet computational constraints in real agricultural scenarios. Sangaiah et al. [23] proposed R-UAV-Net, which combines graph-semantic compression with an enhanced YOLOv4 architecture, thereby reducing computational demands while maintaining high detection accuracy for UAV-based paddy leaf disease monitoring. Similarly, Anandakrishnan et al. [24] developed Tiny-YOLOv9, which incorporates attention and feature adaptation modules to enable efficient, real-time detection on UAVs and edge devices. These studies indicate that optimized, lightweight models can support timely and high-precision crop monitoring under practical deployment constraints.

Previous studies have shown that although convolutional neural networks (CNNs) have achieved significant progress in leaf disease detection, several challenges remain. The common gaps in existing methods are as follows:

1. Recognition accuracy remains suboptimal, particularly in complex and unstructured image backgrounds.

2. CNNs have limited capability to extract fine-grained disease-specific features effectively.

3. Detection performance can be further degraded under adverse weather conditions.

To address these challenges, this study proposes RSA-YOLO, a model based on YOLOv8n [25]. Its lightweight architecture achieves a favorable balance between model size and detection accuracy, making it suitable as a baseline for enhancing detection performance in complex agricultural environments. The proposed RSA-YOLO has the following major contributions:

1. Integration of a Receptive Field Attention (RFA) [26] module to strengthen feature extraction.

2. Replacement of the original Spatial Pyramid Pooling Fast (SPPF) module with the more efficient SPPELAN [27] module to improve feature pyramid representation.

3. Incorporation of an Adaptive Spatial Feature Fusion (ASFF) [28] mechanism in the detection head to optimize cross-scale feature interaction.

In addition, compared with YOLOv8n, RSA-YOLO maintains a controllable computational cost of 11 GFLOPs and a model size of 4.61M parameters, allowing deployment on embedded or resource-constrained agricultural platforms, such as Jetson Nano or mobile edge processors, while significantly improving detection performance.

The rest of the article is organized as follows. Section 2 presents the proposed RSA-YOLO model and its components. Section 3 describes the experimental setup, results, and analysis. Finally, Section 4 concludes the study and discusses future research directions.

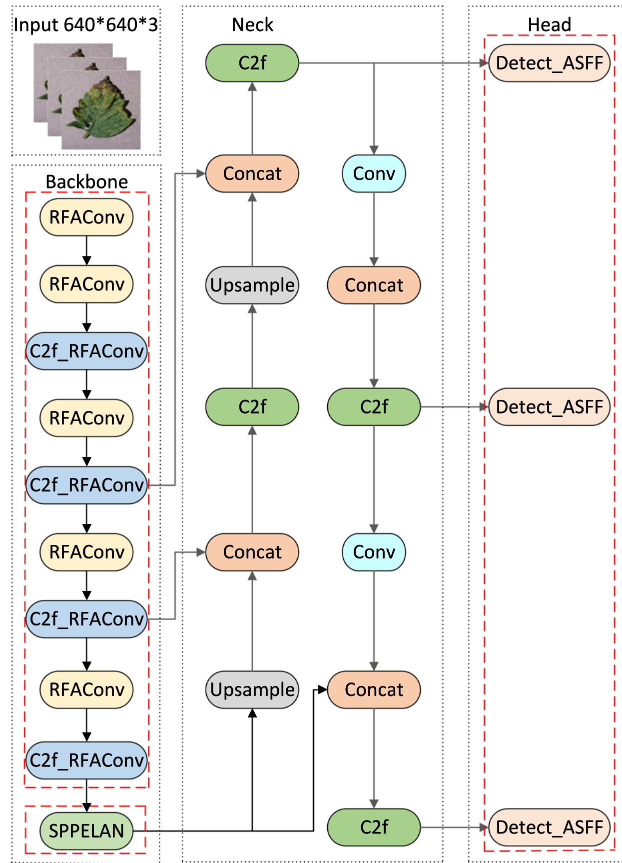


Fig. 1 The structure of RSA-YOLO.

2 Proposed methods

2.1 RSA-YOLO network architecture

YOLOv8 is a single-stage object detection framework evolved from YOLOv5, supporting image classification, object detection, and instance segmentation. Among the YOLOv8 series, YOLOv8n is the smallest model and consists of three main components: the backbone, neck, and detection head. The backbone extracts hierarchical feature representations, which are fused by the neck through multi-scale feature fusion, while the detection head performs classification and bounding box regression.

Based on YOLOv8n, an improved detection model named RSA-YOLO is proposed to enhance the detection accuracy of tomato leaf disease regions under complex backgrounds and multi-scale conditions. To strengthen feature extraction capability, the RFA module is introduced into the backbone to replace the original Conv and C2f modules, improving cross-scale feature representation and robustness under complex scenes. In the neck, the SPPF module is replaced with SPPELAN to enhance multi-scale feature fusion efficiency. Furthermore, the ASFF module is integrated into the detection head to enable adaptive feature fusion across different scales, thereby alleviating scale inconsistency and reducing information loss during multi-scale transmission. The overall network architecture is illustrated in Fig. 1.

2.1.1 Feature enhancement module based on RFA

Traditional convolution operations are prone to local noise interference in complex scenes. This limitation is particularly pronounced under uneven illumination or adverse weather conditions. Such interference can reduce their ability to capture long-range dependencies and contextual information, potentially leading to misdetection or omission. To address this limitation, the RFA module was introduced in this paper, as shown in Fig. 2. Taking the 3×3 convolutional kernel as an example, the “Spatial Feature” represents the initial input feature map, while “Receptive-Field Spatial Feature” refers to the feature mapping generated by the sliding window transformation, consisting of non-overlapping sliding windows. Each receptive-field slider corresponds to a 3×3 window region in the feature map, and these spatial units serve as the basic elements for allocating attention weights.

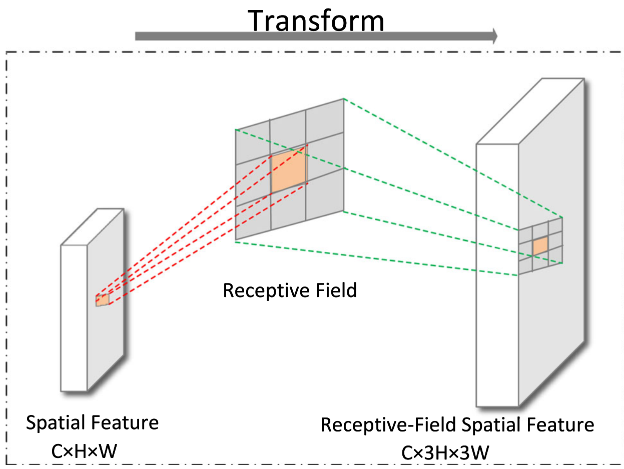


Fig. 2 The receptive-field spatial features are obtained by transforming the spatial features.

Inspired by RFA, we proposed a novel module named RFAConv as a replacement for the standard convolution operation, as illustrated in Fig. 3.The input feature map $X \in R^{C\times H\times W}$ is divided into three groups: Group1, Group2, and Group3, following an average pooling operation. Here, C, H and W represent the channel number, height, and width of the input, respectively. Then, information interaction is achieved via a 1×1 group convolution operation that generates three corresponding convolutional kernels: Kernel1, Kernel2, and Kernel3. Subsequently, an attention map with dimensions $9C \times H \times W$ is produced by applying the Softmax function, emphasizing the importance of each feature within the receptive-field. Meanwhile, the input feature maps undergo a 3×3 group convolution to extract receptive field spatial features with dimensions $9C \times H \times W$. After reshaping, the attention map is fused with the receptive-field spatial features, and a reweighting mechanism is applied to reinforce features in key regions. Through this fusion, the module highlights structure-relevant information and reduces dependence on intensity variations, allowing it to effectively suppress illumination-related noise and stabilize features in fog, shadows, or strong contrast scenes. Finally, the fused features are downsampled using a 3×3 convolution with a stride of 3, and the output preserves the enhanced feature representations with dimensions $C \times H \times W$. The module dynamically assigns weights at the receptive-field level, thereby mitigating the limitations of fixed convolutional kernels while maintaining manageable computational overhead. Based on this dynamic weighting mechanism, the module enhances illumination-invariant feature representation, thereby improving robustness to appearance variations caused by changes in weather and lighting conditions. Furthermore, the module can adaptively focus on key regions and suppress noise, thereby strengthening multi-scale feature representation in complex scenes. The computation of RFA can be formulated as shown in Equation (1):

$$
\begin{aligned}
F &= \mathrm{Softmax}(g^{1\times 1}(\mathrm{AvgPool}(X))) \times \mathrm{ReLU}(\mathrm{Norm}(g^{k\times k}(X))) \\
&= A_{rf} \times F_{rf}
\end{aligned}
\tag{1}
$$

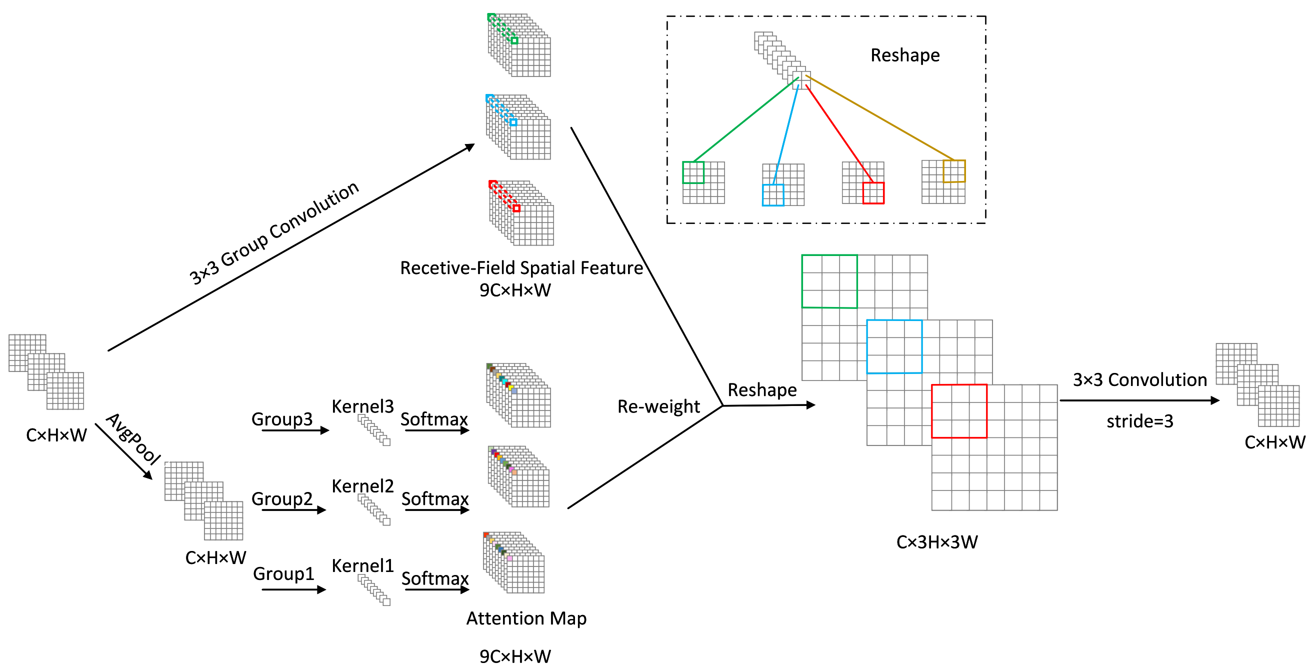


Fig. 3 The structure of RFAConv.

where $g^{i\times i}$ denotes a group convolution of size $i \times i$, k indicates the convolutional kernel size, $Norm$ refers to normalization, and X is the input feature map. F is computed through the multiplication of the attention map A_{rf} and the transformed receptive-field spatial feature F_{rf}. The feature maps obtained by RFA are non-overlapping receptive-field spatial features generated through reshaping, with their attention maps aggregating the feature information of each slider to avoid parameter sharing. In addition, the feature size increases by a factor of k after reshaping, so a $k \times k$ convolutional kernel with a stride of k is applied for feature extraction.

After completing the feature enhancement, the RFAConv mechanism was innovatively integrated into the C2f module of YOLOv8n. The resulting enhancement module retains the benefits of the original residual connections while optimizing feature processing through a partitioning strategy based on receptive-field sliders. This module decouples the input features into multiple sets of receptive-field units for parallel processing. Furthermore, it combines with a deep feature extraction network to enhance the global semantic characterization capability while preserving local detail integrity. This design effectively mitigates the vanishing gradient problem by establishing a cross-level feature interaction mechanism and significantly improves the model's adaptability to variations in target scale through multi-scale feature fusion.

2.1.2 Optimization of SPPELAN structure for multi-scale spatial pyramids

SPPF extracts multi-scale features by performing three cascaded max-pooling operations, which reduces computational complexity. However, since each pooling operation takes the output of the previous one as its input, this cascading process can lead to excessive compression of feature information at certain scales, resulting in the loss of critical details, especially in low-contrast or weather-degraded scenes where textures are weakened. To address this issue, we adopt an SPPELAN-based spatial pyramid structure, in which three independent max-pooling operations are applied in parallel rather than sequentially. Unlike cascaded pooling, each pooling branch in SPPELAN directly operates on the same input feature map, enabling parallel feature extraction across different scales. Even though the pooling kernel size is identical, the independence of parallel paths allows the network to retain complementary feature responses, thereby preventing information loss caused by repeated pooling. The architecture is illustrated in Fig. 4.

In contrast to SPPF, SPPELAN aggregates information from multiple feature maps through independent parallel max-pooling operations. By avoiding inter-stage dependency between pooling operations, this approach improves the capture of multi-level semantic features from the input data and ensures more stable information transfer. The final output of SPPELAN is obtained by concatenating the results from the three parallel max-pooling operations with the original convolutional feature map. The computational procedure is presented in Eqs. (2)–(4).

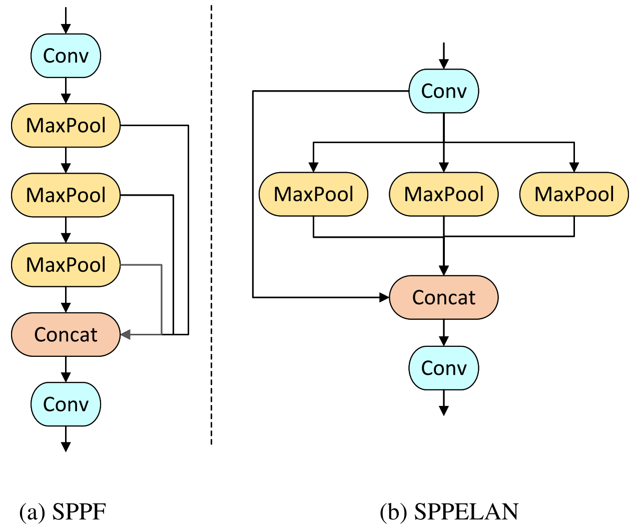


Fig. 4 The structure comparison diagram of SPPF and SPPELAN: a demonstrates the SPPF structure, and b presents the SPPELAN structure.

$$x = \mathrm{Conv}(X_{\mathrm{input}}) \tag{2}$$

$$y_{\max} = \mathrm{Cat}(\mathrm{Max}(x), \mathrm{Max}(x), \mathrm{Max}(x), x) \tag{3}$$

$$y_{\mathrm{out}} = \mathrm{Conv}(y_{\max}) \tag{4}$$

where $Conv$ denotes the convolution operation, Max indicates the max-pooling operation, Cat denotes the concatenation operation along the channel dimension, X_{input} represents the input function, $y_{\max}$ is the output feature of max-pooling, and y_{out} is the final output feature.

By avoiding repeated compression, this parallel pooling mechanism preserves complementary spatial and structural cues, improving robustness under fog, low-light, and shadowed conditions.

2.1.3 Detection head design based on ASFF

The original YOLOv8n detection head employs a fixed feature pyramid structure, enabling multi-scale feature fusion through simple upsampling and concatenation. However, this design may be inadequate when illumination or weather variations differently affect features across scales. Specifically, it lacks cross-scale dynamic adjustment and cannot

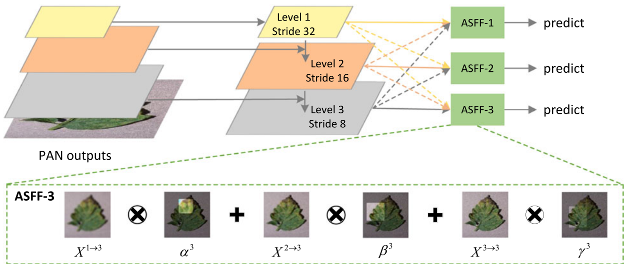


Fig. 5 The structure of ASFF.

fully address scale inconsistencies within the pyramid. As a result, the network provides limited semantic information for small objects and loses spatial detail for large ones, ultimately reducing overall detection performance. To address these issues, we proposed the ASFF module and applied it to the feature maps output by the PAN, thereby constructing an enhanced ASFF-based detection head. ASFF is a data-driven fusion strategy that dynamically adjusts the fusion weights of each scale feature map at every spatial location based on the input content, thereby suppressing conflicting information and enhancing feature representation. It adaptively selects the most relevant features according to object size and location, achieving a balance between semantic richness and spatial detail, and significantly improving multi-scale detection performance.

As illustrated in Fig. 5, which is a schematic illustration of the ASFF feature fusion mechanism, the module acts on three feature map levels (Level 1, Level 2, and Level 3), corresponding to distinct spatial resolutions and semantic levels. ASFF-1, ASFF-2, and ASFF-3 each adaptively fuse feature maps from different levels at their respective scales. The process comprises two key steps:

(1) Feature Resizing: Feature maps from different scales are first resized to the same resolution. Up-sampling: For low-resolution feature maps, the channel dimension is first reduced using a 1×1 convolution and then upscaled via interpolation. Down-sampling: For a down-sampling ratio of $1/2$, a 3×3 convolution with a stride of 2 is applied to simultaneously reduce spatial resolution and adjust the channel dimension. When a larger down-sampling ratio (e.g., $1/4$) is required, max pooling with a stride of 2 is first applied, followed by a convolution with stride 2.

(2) Adaptive Fusion. At each scale l ($l \in \{1, 2, 3\}$), the aligned feature maps are adaptively fused. Specifically, feature maps from other scales are first realigned to match the spatial resolution of scale l, and then fused through a weighted fusion process using learned weights. The fused features at the l-th layer are formulated as shown in Equation (5).

$$y_{ij}^{l} = \alpha_{ij}^{l} \cdot X_{ij}^{1\to l} + \beta_{ij}^{l} \cdot X_{ij}^{2\to l} + \gamma_{ij}^{l} \cdot X_{ij}^{3\to l} \tag{5}$$

where $X_{ij}^{n\to l}$ represents the feature vector at position (i, j) on the feature map aligned from layer n to scale l. y_{ij}^{l} denotes the feature vector of the (i, j)-th channel in the feature map y^{l} after weighted fusion. α_{ij}^{l}, β_{ij}^{l}, and γ_{ij}^{l} are adaptively learned weights representing the importance of the three different scales of the feature map for layer l. α_{ij}^{l}, β_{ij}^{l} and γ_{ij}^{l} can be scalar variables and are consistent across all channels. By enforcing $\alpha_{ij}^{l} + \beta_{ij}^{l} + \gamma_{ij}^{l} = 1$ [29] and ensuring that the values of α_{ij}^{l}, β_{ij}^{l}, and γ_{ij}^{l} range between [0, 1], Equation (6) is defined as follows:

$$\alpha_{ij}^{l} = \frac{e^{\lambda_{\alpha_{ij}}^{l}}}{e^{\lambda_{\alpha_{ij}}^{l}} + e^{\lambda_{\beta_{ij}}^{l}} + e^{\lambda_{\gamma_{ij}}^{l}}} \tag{6}$$

where the weights are computed by the Softmax function. $\lambda_{\alpha_{ij}}^{l}$, $\lambda_{\beta_{ij}}^{l}$ and $\lambda_{\gamma_{ij}}^{l}$ are the key parameters used to control the magnitude of the weights. Through 1×1 convolutional layers, the weight scalar maps λ_{α}^{l}, λ_{β}^{l} and λ_{γ}^{l} are learned from $X_{1\to l}$, $X_{2\to l}$ and $X_{3\to l}$, respectively, and are optimized using standard backpropagation. By spatially predicting these weights adaptively, ASFF can attenuate features degraded by adverse conditions and enhance scales with more stable semantics, thereby improving robustness under varying illumination and environmental conditions.

Using the adaptive aggregation approach described above, features from all layers are adaptively fused at each scale. The resulting output feature maps $\{y_1, y_2, y_3\}$ retain the most relevant information from multiple scales while effectively suppressing conflicting information. These fused feature maps can be directly utilized for object detection tasks, and subsequent processing follows the original YOLOv8n detection pipeline.

3 Experiment and analysis

3.1 Tomato leaf disease dataset

The dataset used in this study was obtained from the publicly available Tomato Leaf Disease dataset on the Roboflow platform, comprising four categories: late blight, septoria leaf spot, leaf miner infestation, and healthy leaves. All images retained their original annotations. The dataset includes images from both natural farmland and laboratory environments. Field samples were collected under varying illumination conditions (morning, noon, and afternoon) and different weather conditions (sunny, cloudy, and rainy), while laboratory samples were acquired under controlled lighting with simple backgrounds. Leaf morphologies include intact, occluded, and overlapping cases, enhancing dataset diversity and supporting model generalization under varied conditions. Although the original dataset provided predefined splits, we

Table 1 Distribution of number of images of tomato leaves

Disease Category	Training Set	Test Set	Validation Set
Late Blight	1493	336	350
Septoria Leaf Spot	1147	228	253
Leaf Miner	3156	643	742
Healthy	5575	856	1255

first repartitioned it into training, validation, and test sets. Data augmentation—including horizontal flipping, brightness adjustment, fog simulation, and random cropping—was then applied exclusively to the training set to further improve sample diversity and enhance model generalization. Brightness adjustment involved randomly scaling contrast by a factor of 0.8–1.2 and shifting brightness by an offset of -20 to 20, while fog simulation added a haze effect with density randomly selected from $[0.03, 0.08]$. After augmentation, the training set contained $3,901$ images (including original and augmented samples), and the validation and test sets each contained 836 images (original images only). This process resulted in an approximately 70%:15%:15% distribution across the three subsets, with class distributions roughly balanced. Detailed distributions of each subset are provided in Table 1.

3.2 Evaluation metrics

To objectively and comprehensively evaluate the detection model's performance on the test set, the following metrics were used in this study: Precision (P), Recall (R), mean Average Precision (mAP), F1-score, Giga Floating Point Operations (GFLOPs), and the number of parameters. P represents the proportion of true positives among all predicted positives, while R represents the proportion of true positives among all actual positives. The F1-score was employed as an evaluation metric to address the issue of class imbalance by providing a harmonic mean of P and R, thereby ensuring a balanced assessment of the model's predictive performance. The specific calculation formulas are provided in Equations (7)–(9).

$$P = \frac{TP}{TP + FP} \times 100\% \tag{7}$$

$$R = \frac{TP}{TP + FN} \times 100\% \tag{8}$$

$$F1 - \text{score} = 2 \times \frac{P \times R}{P + R} \times 100\% \tag{9}$$

GFLOPs are used to measure computational efficiency. The number of parameters reflects the model's complexity and computational cost. mAP is calculated as the mean of average precision scores across all categories and is used to evaluate the overall detection performance of the model. The calculation for Equation (10) is shown below:

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{10}$$

3.3 Implementation details

The experimental setup included an Intel i7-12700H CPU with a 64-bit Windows 11 operating system and an NVIDIA RTX 3050Ti GPU. Additionally, experiments were conducted on a GPU server equipped with an NVIDIA Tesla A40 48GB GPU. The local machine (Windows 11 + RTX 3050Ti) was used for development and training on a subset of the dataset for preliminary experiments, while the Tesla A40 server was employed for training on the full dataset and final evaluation. All algorithms were implemented using the PyTorch deep learning framework in Python. RSA-YOLO was trained on the training dataset with the following hyperparameters: a batch size of 8, 300 training epochs, and an initial learning rate of 0.01. The Stochastic Gradient Descent (SGD) optimizer was employed during training.

3.4 Comparison of different attention

To evaluate the performance of the RFA mechanism, YOLOv8n was selected as the baseline model. Comparative experiments were conducted with the Convolutional Block Attention Module (CBAM) [30], the Similarity-based Attention Module (SimAM) [31], Coordinate Attention (CA), and the Squeeze-and-Excitation (SE) module, assessing their parameter count, computational cost, and detection performance. All attention modules were integrated into the YOLOv8n backbone following their original designs. While the exact insertion positions and number of modules depend on the specific attention mechanism, no additional attention blocks were deliberately stacked or added, ensuring a fair comparison. The results are summarized in Table 2.

The experimental results indicate that RFA-YOLOv8n achieves the highest overall performance, with an mAP@0.5 of 79.3% and an F1-score of 78.5%, outperforming all other attention mechanisms. In contrast, SimAM-YOLOv8n exhibits a decline in performance, with the mAP@0.5 and F1-score decreasing to 77.8% and 77.1%, respectively. Although CBAM-YOLOv8n and CA-YOLOv8n show modest improvements in mAP@0.5 compared with the baseline model, their performance remains inferior to RFA-YOLOv8n, with precision limited to 84%. Moreover, while CBAM-YOLOv8n, CA-YOLOv8n, and SE-YOLOv8n maintain relatively low computational costs, their mAP@0.5 values are still lower than that of RFA-YOLOv8n. Overall,

Table 2 Comparative experiments on various attentional mechanisms

Model	P/%	mAP@0.5/%	F1-score/%	GFLOPs	Params/M
RFA-YOLOv8n	85.6	79.3	78.5	8.7	3.07
SimAM-YOLOv8n	85.3	77.8	77.1	8.9	3.15
CBAM-YOLOv8n	84	78	77.4	8.3	3.11
CA-YOLOv8n	84	78.4	77.9	8.2	3.02
SE-YOLOv8n	85.3	78.1	77.9	8.2	3.01

Table 3 Ablation experiments on each improved module

RFA	ASFF	SPPELAN	P/%	R/%	mAP@0.5/%	F1-score/%	GFLOPs	Params/M
			85.2	70.4	77.9	77.1	8.2	3.01
✓			85.6	72.4	79.3	78.5	8.7	3.07
	✓		85.4	71.9	78.2	78.1	10.4	4.38
		✓	86	71.5	78.4	78.1	8.3	3.17
✓	✓		85.5	73.5	79.7	79.1	10.8	4.44
✓		✓	85.8	71.7	78.8	78.1	8.8	3.23
	✓	✓	86.5	73.2	79.2	79.3	10.5	4.54
✓	✓	✓	86.8	73.3	80.5	79.5	11	4.61

RFA-YOLOv8n achieves a more favorable trade-off between detection accuracy and computational efficiency, representing the most balanced model among the evaluated attention mechanisms.

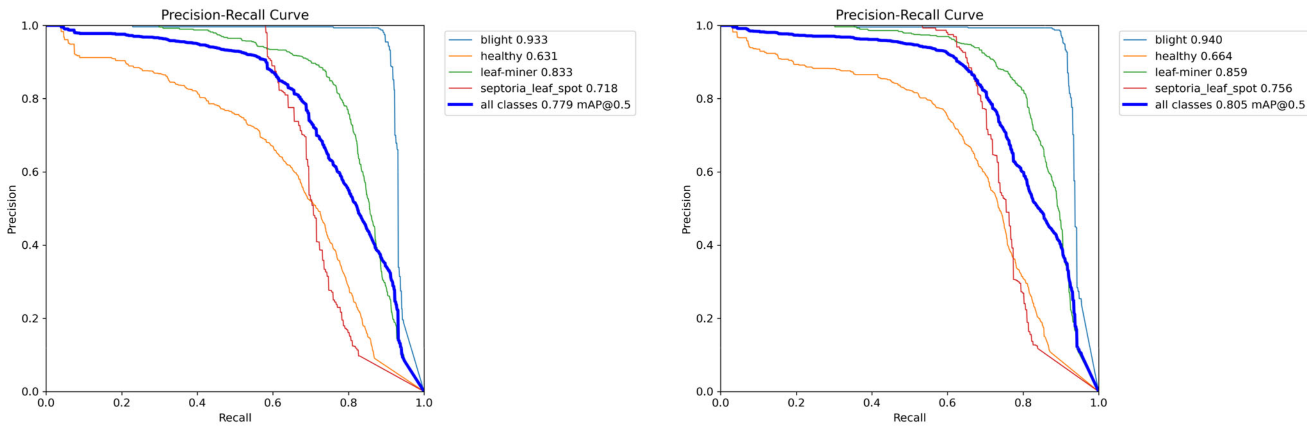
3.5 Ablation experiments

To validate the effectiveness of the proposed RSA-YOLO model for tomato leaf disease detection, ablation experiments were conducted based on the original YOLOv8n, focusing on three key components: the RFA, ASFF, and SPPELAN modules. Precision, recall, mAP@0.5, F1-score, GFLOPs, and the number of parameters were adopted as evaluation metrics, and the results of different module configurations are summarized in Table 3. As shown in Table 3, first introducing the RFA module leads to a noticeable improvement in detection performance, with recall increasing by 2.0% and both mAP@0.5 and F1-score improving by 1.4%, indicating that dynamic receptive-field modeling effectively enhances the localization ability for disease targets. When the ASFF module is further incorporated, the detection accuracy is further improved, and the mAP@0.5 increases to 79.7%, demonstrating that adaptive multi-scale feature fusion strengthens the representation of subtle lesion features. Subsequently, introducing the SPPELAN module brings additional improvements in precision and mAP@0.5, confirming its effectiveness in refining the feature extraction process. When all three modules are integrated simultaneously, the model achieves the best overall performance, with precision, recall, mAP@0.5, and F1-score reaching 86.8%, 73.3%, 80.5%, and 79.5%, respectively. Overall, the progressive integration of each module consistently enhances detection accuracy, and despite a moderate increase in computational complexity, the final RSA-YOLO model maintains a favorable balance between detection performance and computational cost.

Fig. 6 presents the precision–recall (PR) curves of the baseline YOLOv8n model and the proposed RSA-YOLO on the tomato leaf disease dataset. Overall, RSA-YOLO outperforms YOLOv8n across most recall levels, with an increased area under the PR curve, indicating improved accuracy and robustness for detecting different diseases. This result further confirms the effectiveness of the proposed enhancement modules.

3.6 Analysis of experimental results on different datasets

To further evaluate the performance and generalization capability of the proposed RSA-YOLO model, we conducted additional experiments on an external dataset, referred to as Dataset I. This dataset, obtained from the Kaggle platform (Tomato Leaf Diseases Detection Computer Vision), is publicly available and was used without any modifications or extensions. It contains seven categories, including several additional disease categories that are absent from the Tomato Leaf Disease dataset used for the main experiments. Table 4 summarizes the evaluation results in terms of detection performance and computational cost, including P, mAP@0.5, F1-score, GFLOPs, and model parameter count. Compared with the baseline YOLOv8n, RSA-YOLO improved P, mAP@0.5, and F1-score by 5.2%, 1.4%, and 2%, respectively. Although the computational cost (GFLOPs) increased from 8.2 to 11 and the number of parameters grew



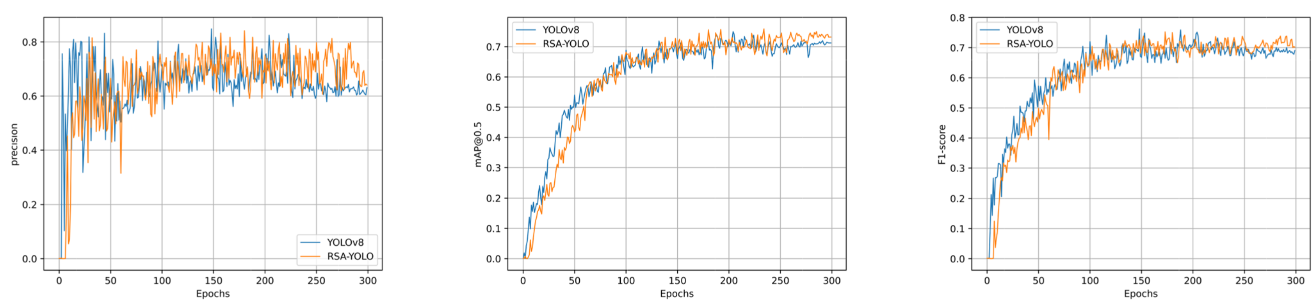
(a) YOLOv8n

(b) RSA-YOLO

Fig. 6 Comparison of Precision–Recall performance between YOLOv8n and RSA-YOLO.

Table 4 Comparison of Dataset I before and after improvement

Dataset I	P/%	mAP@0.5/%	F1-score/%	GFLOPs	Params/M
YOLOv8n	65.2	72.7	70.9	8.2	3.01
RSA-YOLO (ours)	70.4	74.1	72.9	11	4.61



(a) P

(b) mAP@0.5

(c) F1-score

Fig. 7 Comparison of model training results on Dataset I.

from 3.01 M to 4.61 M, the observed improvements demonstrate the effectiveness of the proposed modifications. These results indicate that RSA-YOLO not only improves detection performance for small or challenging objects but also demonstrates robust generalization to novel categories and previously unseen samples, confirming its adaptability under diverse data conditions.

As shown in Fig. 7, the training curves on Dataset I demonstrate that RSA-YOLO consistently outperforms the baseline YOLOv8n across multiple evaluation metrics. In terms of precision, RSA-YOLO exhibits a more stable upward trend after the initial fluctuations, indicating stronger robustness in distinguishing positive samples. For mAP@0.5, both models improve rapidly during the early epochs; however, RSA-YOLO reaches a higher convergence level and maintains a more stable plateau in the later stages, suggesting superior overall detection capability. With respect to the F1-score, although the two models follow similar trajectories, RSA-YOLO achieves slightly higher values across most epochs, reflecting a better balance between precision and recall. Collectively, these results confirm that the proposed improvements effectively enhance the model's generalization performance on a previously unseen dataset.

3.7 Comparison of different algorithms

To validate the effectiveness of the proposed RSA-YOLO model, we conducted comparative experiments with several mainstream object detection algorithms under identical conditions, including YOLOv3-tiny, YOLOv5n, YOLOv6n, YOLOv8n, YOLOv9C, YOLOv11n, YOLOv12n and YOLOv13n. All models were trained on the tomato leaf disease dataset using uniform settings and evaluated on the

Table 5 Comparative experiments of various mainstream target detection models

Model	P/%	mAP@0.5/%	F1-score/%	GFLOPs	Params/M
YOLOv3-tiny	79.5	73.7	73.5	18.9	12.13
YOLOv5n	82.3	76.4	76.2	7.3	4.13
YOLOv6n	86.3	76	76	11.8	4.23
YOLOv8n	85.2	77.9	77.1	8.2	3.01
YOLOv9C	84.8	78.1	77.9	118.2	31.4
YOLOv11n	85	77.9	78.5	6.4	2.5
YOLOv12n	85.1	77	78.2	6	2.52
YOLOv13n	85.9	78.6	78.5	6.3	2.46
RSA-YOLO (ours)	86.8	80.5	79.5	11	4.61

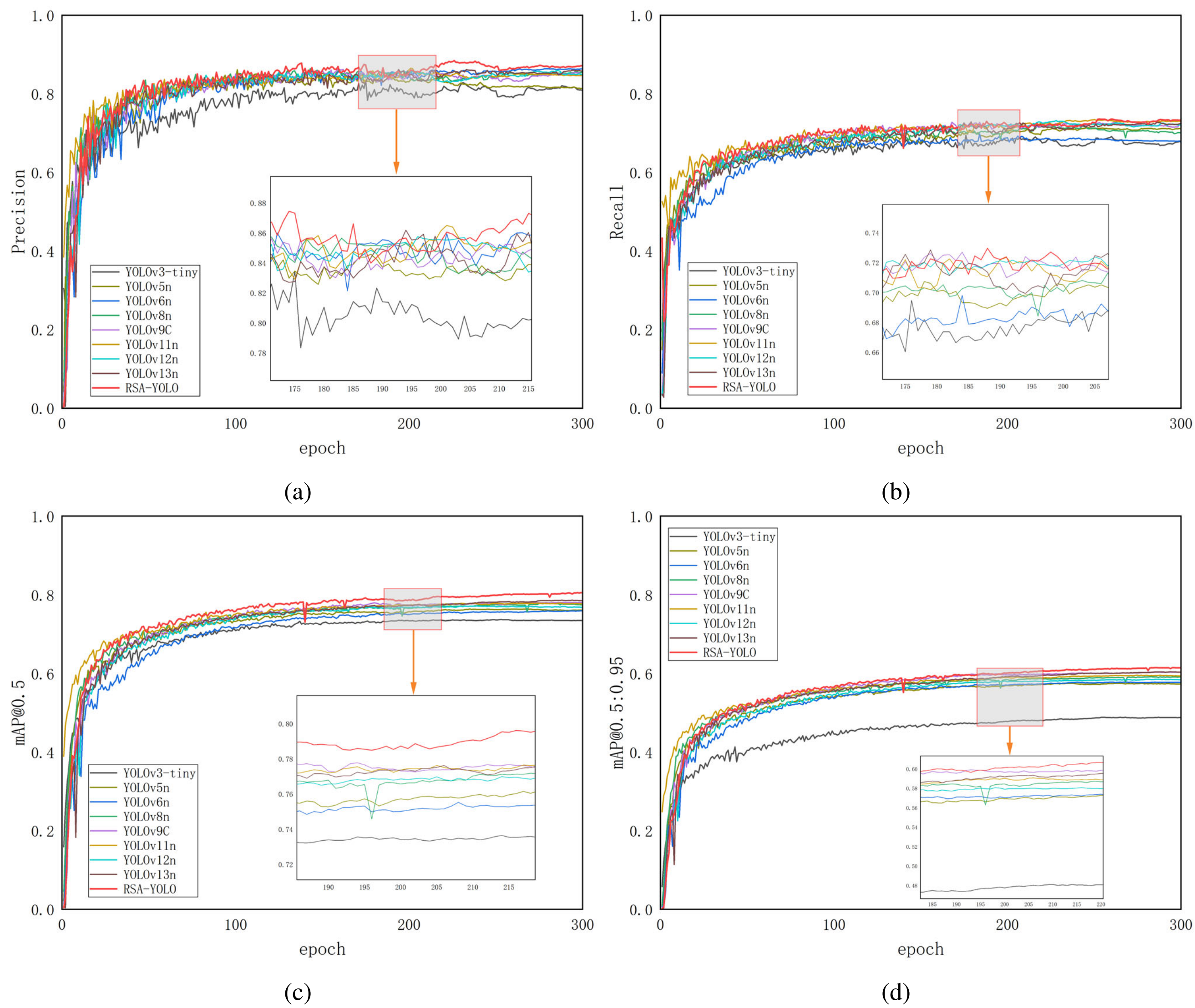


Fig. 8 The line plots of Precision, Recall, mAP@0.5 and mAP@0.5: 0.95 for different models: a Precision; b Recall; c mAP@0.5; d mAP@0.5: 0.95.

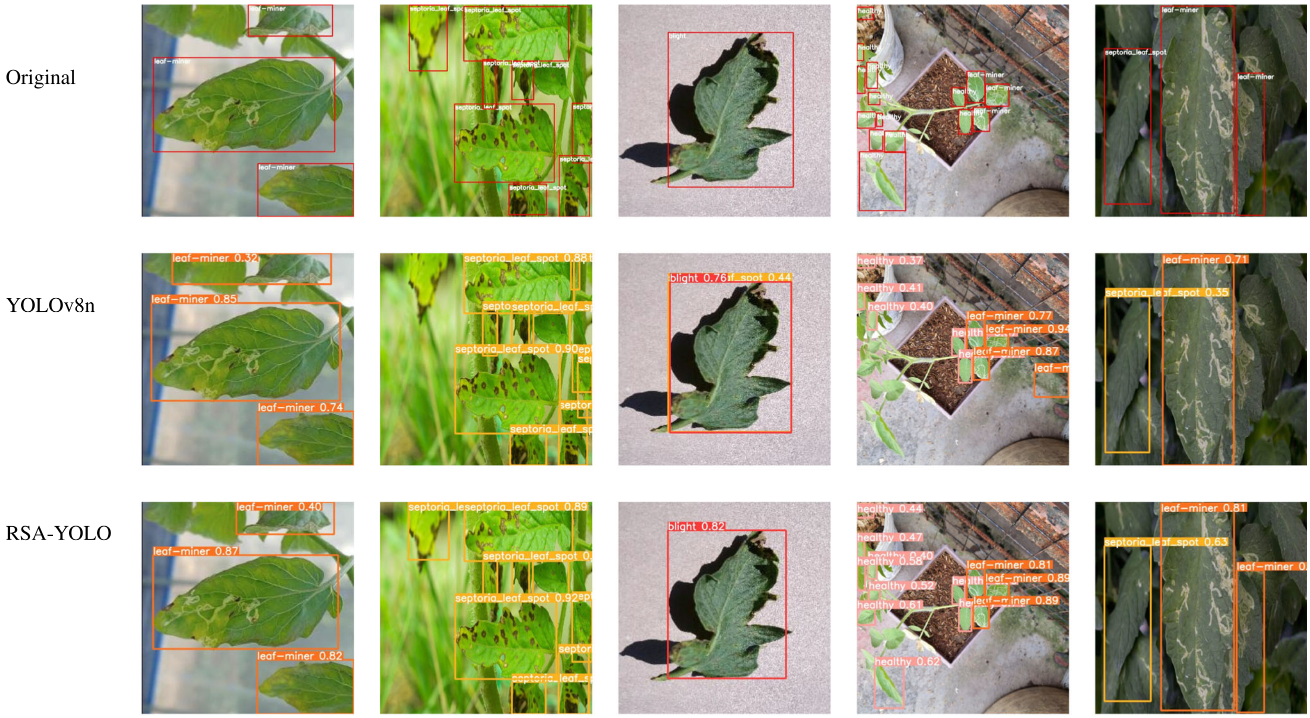


Fig. 9 Comparison between YOLOv8n and RSA-YOLO in tomato leaf disease detection results.

corresponding test set. The results are summarized in Table 5.

Table 5 shows that RSA-YOLO achieves superior performance across the main evaluation metrics, with a precision of 86.8%, mAP@0.5 of 80.5%, and an F1-score of 79.5%, outperforming the other models. Compared with the lightweight YOLOv13n (2.46 M parameters), RSA-YOLO provides higher detection accuracy while maintaining moderate computational costs (11 GFLOPs, 4.61 M parameters). In addition, compared with YOLOv9C, which incurs substantially higher computational overhead (118.2 GFLOPs, 31.4 M parameters), RSA-YOLO attains a more favorable trade-off between accuracy and complexity. These findings suggest that RSA-YOLO not only improves detection accuracy over existing mainstream models but also offers a balanced compromise between model size and computational efficiency, underscoring its potential for practical deployment in agricultural applications.

Using detection precision, recall, mAP@0.5, and mAP@0.5:0.95 as evaluation metrics, the performance curves for the different models on the test set were plotted, as shown in Fig. 8. During the initial phase of training, the curves for all evaluation metrics across all models increased rapidly, which indicated effective learning of data features. As training progressed, RSA-YOLO demonstrated significant advantages in detection precision, mAP@0.5, and mAP@0.5:0.95, all of which showed considerably higher values than the other models. Notably, the recall curve of RSA-YOLO remained high, reflecting its ability to detect more true objects and substantially reduce the miss rate. Furthermore, RSA-YOLO not only converged faster but also exhibited smaller fluctuations in the evaluation curves of metrics during the later stages of training, indicating improved generalization and training stability.

3.8 Visualization and analysis of results

To clearly demonstrate the effectiveness of RSA-YOLO, images of different tomato leaf diseases were used as input. These images were detected using both the original YOLOv8n and RSA-YOLO models, and the corresponding visualization results are presented in Fig. 9.

As illustrated in the figure, compared with the original YOLOv8n model, RSA-YOLO not only generates bounding boxes that more accurately correspond to the actual target sizes, but also substantially reduces both missed detections and false positives, indicating improved localization accuracy and enhanced detection reliability for tomato leaf diseases.

4 Conclusion

In this study, to address the challenges posed by the varying sizes of tomato leaf diseases and complex backgrounds, we proposed a novel model named RSA-YOLO. By integrating the RFA module into the backbone of YOLOv8n, the model more effectively captured fine-grained details of small objects and semantic features of larger ones, thereby enhancing multi-scale feature extraction. To further enhance the representational power of the feature pyramid, the original SPPF module in the backbone was replaced with the SPPELAN module, resulting in improved efficiency and accuracy in multi-scale feature fusion. Additionally, the ASFF module was incorporated into the detection head to ensure scale invariance of the enhanced features and further boost detection accuracy. Experimental results demonstrated that RSA-YOLO significantly outperformed the original YOLOv8n, achieving a 1.6% increase in precision, a 2.6% improvement in mAP@0.5, and a 2.4% gain in F1-score. Compared to YOLOv3-tiny, YOLOv5n, YOLOv6n, YOLOv9C, YOLOv11n, YOLOv12n and YOLOv13n, RSA-YOLO achieved higher mAP@0.5 values by 6.8%, 4.1%, 4.5%, 2.4%, 2.6%, 3.5%, and 1.9%, respectively. These results indicate that RSA-YOLO offers a more effective approach for the early detection of tomato leaf disease.

Although the RSA-YOLO model achieves a significant improvement in detection accuracy and effectively alleviates the challenges posed by multi-scale blurred targets, background interference, and illumination variations, its parameter count is slightly higher than that of the baseline model. Nevertheless, this increase remains within a reasonable range and has little impact on overall performance. Meanwhile, although missed detections have been substantially reduced, some healthy leaves may still be overlooked, as illustrated in Fig. 9. These cases typically occur when leaves exhibit abnormal morphology, severe overlap, or strong similarity to the surrounding environment, revealing the model's limitations under extreme and complex conditions. Future work will focus on further enhancing deployment performance by applying model compression techniques such as pruning and quantization, thereby reducing model size while maintaining accuracy. These efforts aim to improve efficiency and practicality, facilitating real-time disease detection in field applications on smartphones or portable agricultural devices.

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Declarations

Conflicts of Interest The authors declare no competing interests.

References

1. Xu, Z., Shou, W., Huang, K., et al.: The current situation and trend of tomato cultivation in China. Acta Physiol. Plant. **22**, 379–382 (2000). https://doi.org/10.1007/s11738-000-0061-y
2. Fuentes, A., Yoon, S., Youngki, H., Lee, Y., Park, D. S.: Characteristics of tomato plant diseases—a study for tomato plant disease identification. In: *Proc Int Symp Inf Technol Converg*, vol. 1, pp. 226-231(2016)
3. Wang, X., Liu, J., Zhu, X.: Early real-time detection algorithm of tomato diseases and pests in the natural environment. Plant Methods **17**, 43 (2021). https://doi.org/10.1186/s13007-021-00745-2
4. Yao, J., Tran, S.N., Sawyer, S., et al.: Machine learning for leaf disease classification: data, techniques and applications. Artif. Intell. Rev. **56**(Suppl 3), 3571–3616 (2023). https://doi.org/10.1007/s10462-023-10610-4
5. Ullah, N., Khan, J.A., Alharbi, L.A., et al.: An efficient approach for crops pests recognition and classification based on novel deeppestnet deep learning model. IEEE Access **10**, 73019–73032 (2022). https://doi.org/10.1109/ACCESS.2022.3189676
6. Liu, J., Wang, X.: Plant diseases and pests detection based on deep learning: a review. Plant Methods **17**, 22 (2021). https://doi.org/10.1186/s13007-021-00722-9
7. Madhav, M.U., Jyothi, D.N., Kalyani, P.: Prediction of pesticides and identification of diseases in fruits using Support Vector Machine (SVM) and IoT. In: *AIP Conference Proceedings*, vol. 2407, no. 1(2021) https://doi.org/10.1063/5.0074595
8. Bera, A., Bhattacharjee, D., Krejcar, O.: PND-Net: plant nutrition deficiency and disease classification using graph convolutional network. Sci. Rep. **14**, 15537 (2024). https://doi.org/10.1038/s41598-024-66543-7
9. Prabu, M., Chelliah, B.J.: An intelligent approach using boosted support vector machine based arithmetic optimization algorithm for accurate detection of plant leaf disease. Patt. Anal. Appl. **26**, 367–379 (2023). https://doi.org/10.1007/s10044-022-01086-z
10. Wu, J., Abolghasemi, V., Anisi, M.H., et al.: Strawberry disease detection through an advanced squeeze-and-excitation deep learning model. *IEEE Transactions on AgriFood Electronics*(2024)
11. Iqbal, I., Odesanmi, G.A., Wang, J., et al.: Comparative investigation of learning algorithms for image classification with small dataset. Appl. Artif. Intell. **35**(10), 697–716 (2021)
12. Girshick, R., Donahue, J., Darrell, T., et al.: Rich feature hierarchies for accurate object detection and semantic segmentation. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 580–587(2014)
13. Ren, S., He, K., Girshick, R., et al.: Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems* **28**(2015)
14. Zhang, H., Wang, Y., Dayoub, F., et al.: Varifocalnet: An iou-aware dense object detector. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8514–8523(2021)
15. Hussain, M.: Yolov1 to v8: Unveiling each variant–a comprehensive review of yolo. IEEE Access **12**, 42816–42833 (2024)
16. Ge, Z., Liu, S., Wang, F., et al.: YOLOX: Exceeding YOLO series in 2021.(2021) https://doi.org/10.48550/arXiv.2107.08430
17. Sun, H., Xu, H., Liu, B., He, D., He, J., Zhang, H., Geng, N.: MEAN-SSD: A novel real-time detector for apple leaf diseases using improved light-weight convolutional neural networks. Comput. Electron. Agric. **189**, 106379 (2021). https://doi.org/10.1016/j.compag.2021.106379
18. Ullah, Z., Alsubaie, N., Jamjoom, M., Alajmani, S.H., Saleem, F.: EffiMob-Net: A Deep Learning-Based Hybrid Model for Detection and Identification of Tomato Diseases Using Leaf Images. Agriculture **13**, 737 (2023). https://doi.org/10.3390/agriculture13030737

19. Zhang, R., Ge, Y., Li, Q., et al.: An Improved YOLOv8 Tomato Leaf Disease Detector Based on the Efficient-Net backbone. In: *Proceedings of the 6th International Symposium on Advanced Technologies and Applications in the Internet of Things*, Shiga, Japan, pp. 19–22(2024)
20. Chen, M., Wang, C., Liu, C., et al.: Tomato leaf disease detection method based on improved YOLOv8n. *Scientific Reports* **15**(1), 25837(2025) https://doi.org/10.1038/s41598-025-00405-8
21. Miao, Y., Meng, W., Zhou, X.: SerpensGate-YOLOv8: an enhanced YOLOv8 model for accurate plant disease detection. Front. Plant Sci. **15**, 1514832 (2025). https://doi.org/10.3389/fpls.2024.1514832
22. Qiang, J., Liu, W., Li, X., Guan, P., Du, Y., Liu, B., Xiao, G.: Detection of citrus pests in double backbone network based on single shot multibox detector. Comput. Electron. Agric. **212**, 108158 (2023). https://doi.org/10.1016/j.compag.2023.108158
23. Sangaiah, A.K., Anandakrishnan, J., Devarapelly, A.R., et al.: R-UAV-Net: Enhanced YOLOv4 With Graph-Semantic Compression for Transformative UAV Sensing in Paddy Agronomy. IEEE Transactions on Cognitive Communications and Networking **11**(2), 1197–1209 (2025). https://doi.org/10.1109/TCCN.2024.3452053
24. Anandakrishnan, J., Sangaiah, A. K., Son, N. K., et al.: UAV-Based Deep Learning with Tiny-YOLOv9 for Revolutionizing Paddy Rice Disease Detection. In: *Proceedings of the 2024 IEEE International Conference on Smart Internet of Things (SmartIoT)*, Shenzhen, China, pp. 16–21 (2024) https://doi.org/10.1109/SmartIoT62235.2024.00012
25. Yaseen, M.: What is YOLOv8: An in-depth exploration of the internal features of the next-generation object detector.(2024)
26. Zhang, X., Liu, C., Yang, D., et al.: RFAConv: Innovating spatial attention and standard convolutional operation.(2023) https://doi.org/10.48550/arXiv.2304.03198
27. Wang, C.Y., Yeh, I.H., Mark, Liao, H.Y.: Yolov9: Learning what you want to learn using programmable gradient information. In: *European conference on computer vision*, pp. 1–21(2024) https://doi.org/10.1007/978-3-031-72751-1_1
28. Liu, S., Huang, D., Wang, Y.: Learning spatial fusion for single-shot object detection.(2019) https://doi.org/10.48550/arXiv.1911.09516
29. Wang, G., Wang, K., Lin, L.: Adaptively connected neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 1781-1790(2019)
30. Woo, S., Park, J., Lee, J.Y., et al.: Cbam: Convolutional block attention module. In *Proceedings of the European conference on computer vision (ECCV)*, pp. 3-19(2018)
31. Yang, L., Zhang, R.Y., Li, L., et al.: Simam: A simple, parameter-free attention module for convolutional neural networks. In *International conference on machine learning*, pp. 11863-11874(2021)